Towards greener and miniaturized paprika metabolomics: LC-HRMS-based multiclass profiling strategy for cultivar discrimination

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Escanéame

CONTACT

Postdoctoral researcher
Araceli Rivera Pérez
arp800@ual.es

UNIVERSIDAD DE ALMERÍA · QAC

Araceli Rivera-Pérez[a,b], Antonia Garrido Frenich[a]

[a]Research Group "Analytical Chemistry of Contaminants", Department of Chemistry and Physics, CIAIMBITAL, ceiA3, University of Almeria, Almeria, E-04120 Almeria, (Spain)

[b]Department of Analytical Chemistry, Faculty of Sciences, University of Granada, Campus Fuentenueva s/n, E-18071, Granada, (Spain)

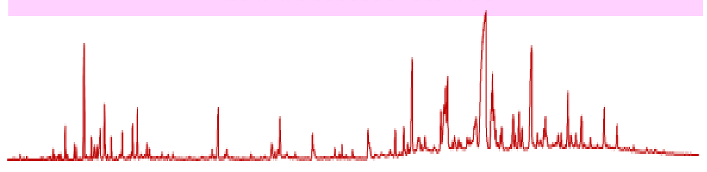
arp800@ual.es

1. INTRODUCTION

- Green metabolomics is gaining increasing attention as analytical workflows strive to reduce solvent consumption, sample requirements and environmental impact. In this context, paprika (*Capsicum annuum* L.), a high-value food product rich in carotenoids, capsaicinoids and phenolics, represents an ideal model for the development of more sustainable metabolomics strategies.
- Since the chemical characterization of many paprika cultivars remains limited, this study aimed to develop a green and miniaturized UHPLC-Q-Orbitrap-HRMS metabolomics workflow for the characterization and differentiation of two previously unexplored Chinese paprika cultivars (Hong Long 23 and ZYH 30).
- The proposed strategy combined µ-UAE extraction, suspect screening, multivariate analysis and QUANTEM-based semi-quantification, enabling cultivar discrimination, biomarker discovery and reduced dependence on analytical standards.

2. AIM OF STUDY AND APPLIED APPROACH

COMPARATIVE METABOLOMICS
for Chinese cultivar discrimination

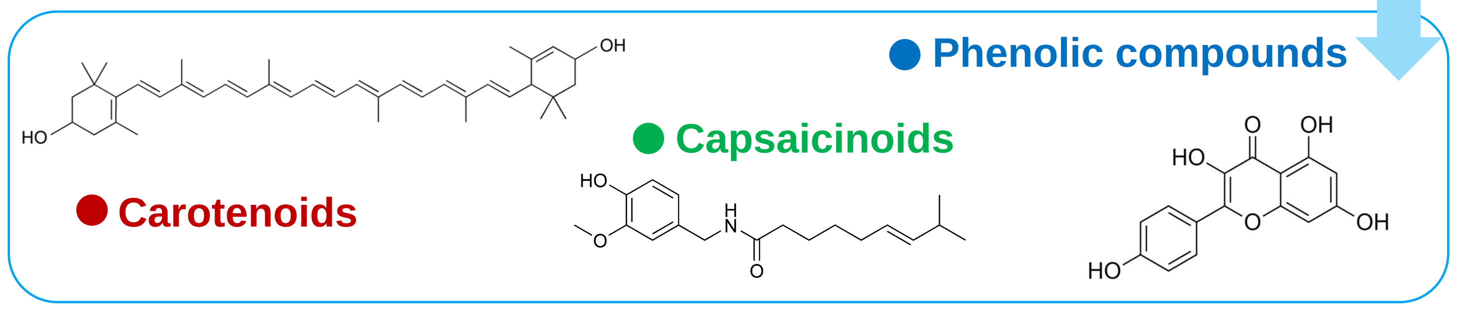


MULTICLASS METABOLITE PROFILING
for analytical coverage of key paprika's chemicals

Hong Long 23 (n= 30)

ZYH 30 (n= 30)

UHPLC-Q-Orbitrap-HRMS



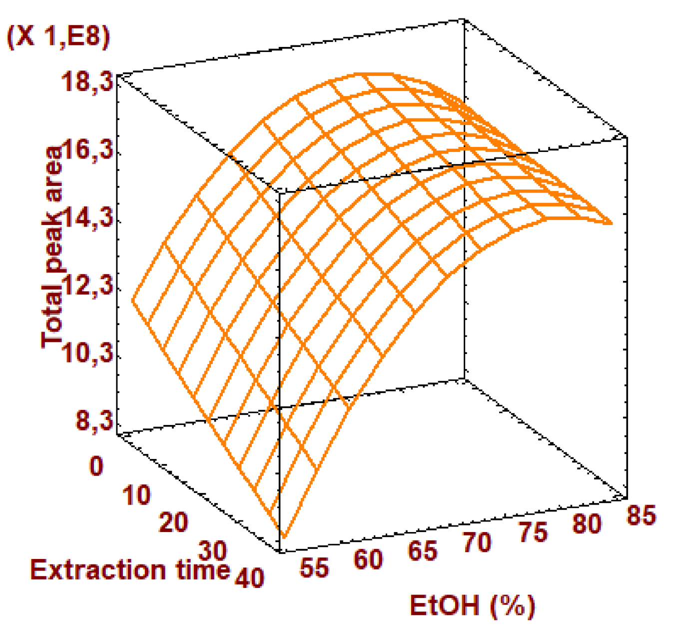
3. GREEN AND MINIATURIZED SAMPLE EXTRACTION FOR MULTICLASS PAPRIKA PROFILING BY UHPLC-HRMS ANALYSIS

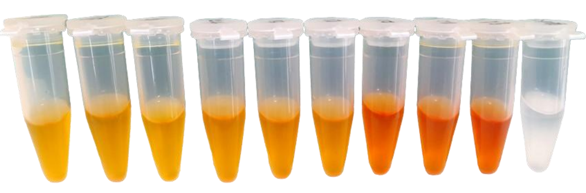
3.1. DESIGN OF EXPERIMENT (DOE) OPTIMIZATION

Response Surface Methodology (RSM) statgraphics 19 centurion

Factor 1
Ethanol (%)
60-80% (*v/v*)

Factor 2
Extraction time (min)
10-30 min





2-factor Central Composite Design (CCD)

Optimal selected conditions
based on maximum overall response using peak areas of 9 representative metabolites

- EtOH 75% (*v/v*)
- Only 6 extraction minutes

3.2. EXTRACTION WORKFLOW

1. SAMPLE WEIGHING

40 mg of paprika powder

2. µ-UAE EXTRACTION

6 min with 0.8 mL of EtOH:water (75:25 v/v) 0.05% 3-BHA (antioxidant)

3. CENTRIFUGATION AND FILTRATION

10 min (12000 rpm, 10 ºC) PTFE filters 0.2 µm

4. SAMPLE DILUTION AND UHPLC-HRMS ANALYSIS

1:4 (v/v) with the extraction solvent

3.3. PROFILING OF PAPRIKA SAMPLES

1. RAW FINGERPRINTING DATA
Full scan and dd-MS² data ESI (+) mode

2. SUSPECT SCREENING
with Compound Discoverer™ software using a targeted list
In-house carotenoid and capsaicinoid database and *Phenol-Explorer 3.6 database*

3. FEATURE ANNOTATION
- *Mass errors < 5 ppm*
- *MS and MS/MS data matching with identification tools*
- *Confidence Level: 2a-b*

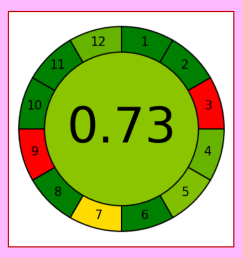
Level 2a Library spectrum match
Level 2b In silico/predicted match

Experimental MS/MS libraries (e.g. mzCloud, MassBank, FooDB)
In silico tools (e.g. CFM-ID, Sirius)

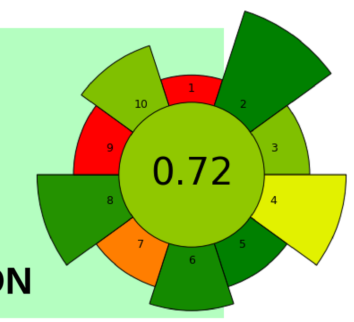
38 compounds putatively identified for cultivar differentiation

4. METHOD METRICS

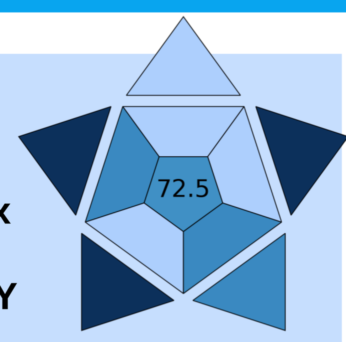
AGREE
Analytical GREEnness Metric
GREEN ANALYTICAL METHOD



AGREEprep
Analytical GREEnness Metric for Sample PREParation
SUSTAINABLE SAMPLE PREPARATION

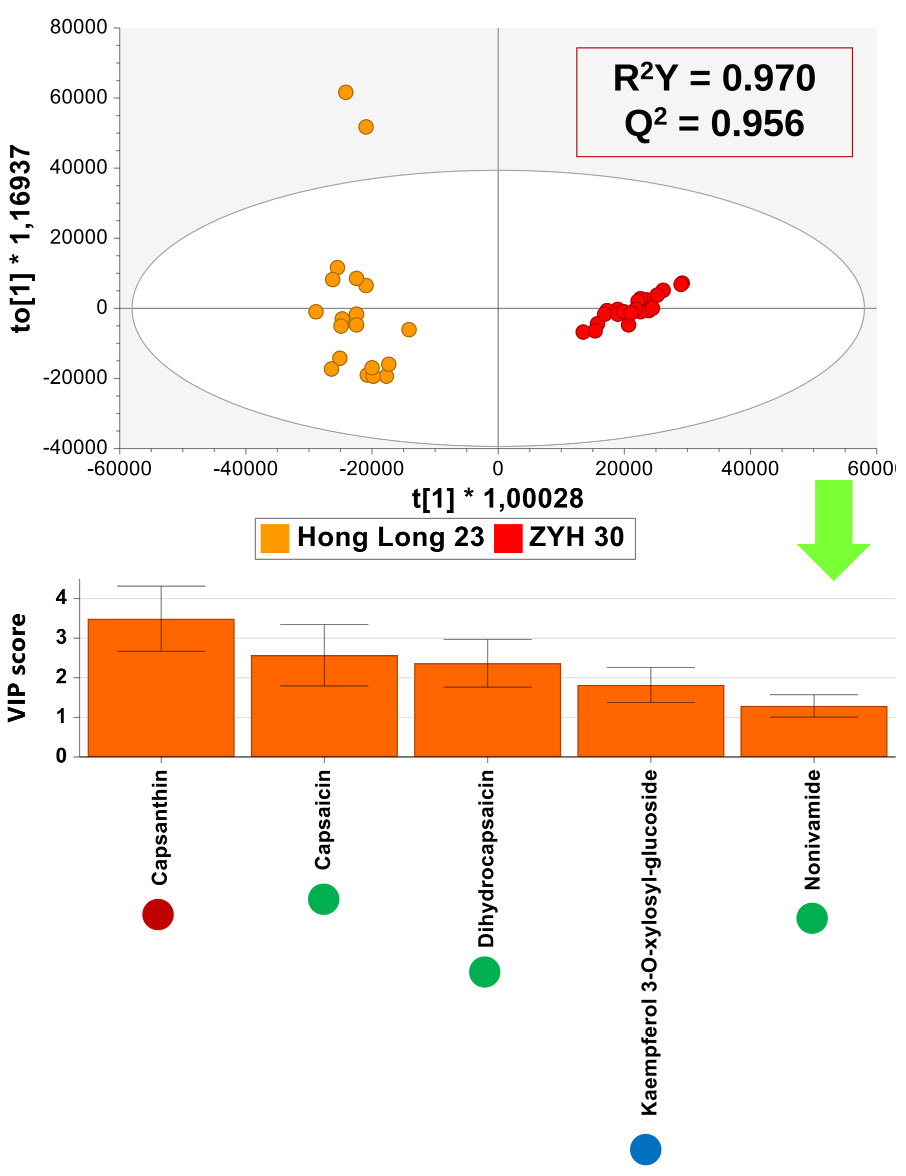


BAGI
Blue Applicability Grade Index
HIGH METHOD APPLICABILITY



5. MULTIVARIATE DATA ANALYSIS AND MARKER IDENTIFICATION

5.1. Orthogonal Partial Least Squares Discriminant Analysis (OPLS-DA)



5.2. Identification of discriminant and significant metabolites

Marker selection workflow

Variable Importance in Projection (VIP) score > 1.00

Benjamini–Hochberg-adjusted *p*-values < 0.05

Fold change (FC) cut-off > 1.10

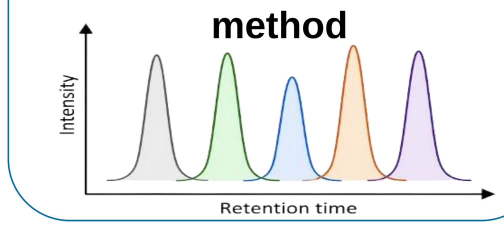
Two characteristic ions (precursor and at least one fragment ion) with mass accuracy ≤ 5 ppm

5 compounds for reliable cultivar discrimination

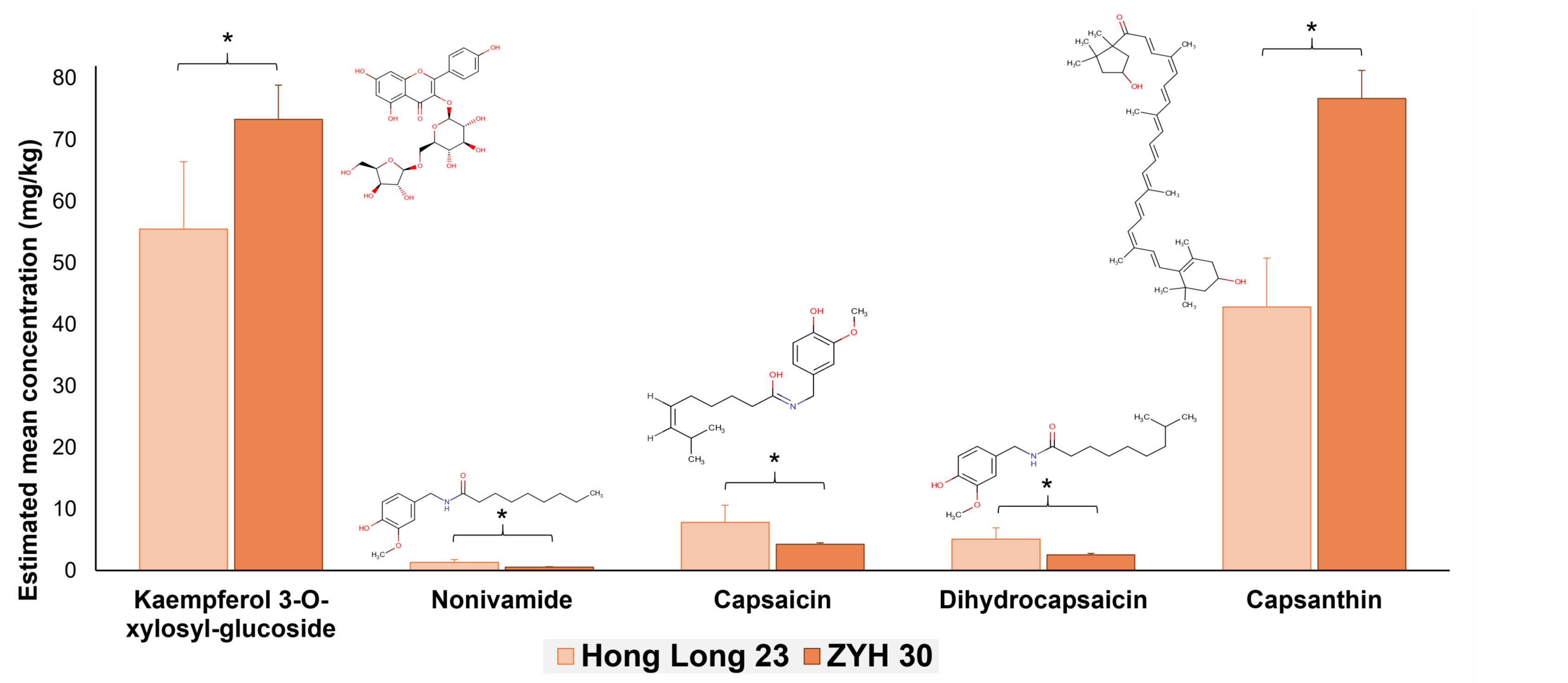
6. TOWARDS GREEN METABOLOMICS USING STANDARD-FREE SEMI-QUANTIFICATION OF VIP MARKERS

Quantem ANALYTICS
Relative quantification of markers using Quantem approach for suspect screening

5 tailoring compounds with known concentration measured with the same method



TAILORING COMPOUND SELECTION CRITERIA
- RT coverage across the chromatogram
- High MS response and stability in ESI (+)
- Not present in samples



Semi-quantification of discriminant metabolites using Quantem and tailored surrogate compounds avoided the need for expensive, non-commercially available carotenoid or capsaicinoid authentic standards, in line with greener metabolomics workflows [1]

7. CONCLUSIONS

✓ Untargeted metabolomics successfully discriminated sample groups and revealed key chemical markers.

✓ Five VIP metabolites were identified as the main drivers of sample differentiation.

✓ Semi-quantification was achieved using Quantem and tailored surrogate compounds, avoiding the need for authentic standards.

✓ The proposed workflow combines analytical performance with enhanced sustainability, supporting greener metabolomics studies.

Take-home message: Green metabolomics is feasible: reliable biomarker discovery and semi-quantification can be achieved while minimizing the use of analytical standards and resources.

8. ACKNOWLEDGMENTS

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Junta de Andalucía · Unión Europea Fondo Europeo de Desarrollo Regional · Andalucía se mueve con Europa

9. REFERENCES

[1] C. Spaggiari, K. Othibeng, F. Tugizimana, G. Rocchetti, L. Righetti, Towards a greener future: *The role of sustainable methodologies in metabolomics research*, Adv. Sample Prep. 14 (2025) 100186.